

Spatial orientation: Model-based approach to multi-sensory mechanisms

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Abstract

Perception of spatial orientation is generated through multimodal sensory integration. In this process, there are systematic errors with changes in the head or body position, which reflect challenges for the brain in maintaining a common sensory reference frame for spatial orientation. Here, we focus on this multisensory aspect of spatial orientation. We review a Bayesian spatial perception model that can be used as a framework to study sensory contributions to spatial orientation during lateral head tilts and probe neural networks involved in this process.

Keywords

Spatial orientation, SVV, Subjective visual vertical, Sensory integration, Bayesian

1 Introduction

As a fundamental reference for orientation in space, gravity is critical in human balance, perception, and behavior (Clark et al., 2015; Clément and Eckardt, 2005; Jenkin et al., 2004; Lipshits et al., 2005; Lopez et al., 2009; Mast, 2000; Mikellidou et al., 2015). Aligned with the direction of gravity, our brain maintains a stable perception of the world despite frequent changes in the eye, head and body positions. Such perceptual stability is known as orientation constancy, and when disrupted, there are often debilitating consequences due to ensuing spatial disorientation, dizziness and imbalance (Kheradmand and Winnick, 2017). In the process of

maintaining spatial orientation, the compensatory vestibulo-ocular response is critical for visual stability during changes in the head position. In humans, however, lateral head tilts (i.e., with respect to gravity), as opposed to the horizontal and vertical head movements, lead to changes in the torsional eye position that do not match the magnitude of the head tilt (Leigh and Zee, 2015). This partial change in torsional eye position, referred to as ocular counter-roll (OCR), results in a visual-vestibular mismatch, which requires the brain to integrate sensory signals that are encoded in different reference frames in order to maintain a coherent spatial perception. This multisensory integration involves graviceptive signals from the otoliths, visual inputs from the retina, and other sensory inputs that encode the eye, head, and body positions (Fig. 1). Here, we focus on this multisensory aspect of spatial orientation during a static lateral head tilt. We review a spatial perception model as a framework

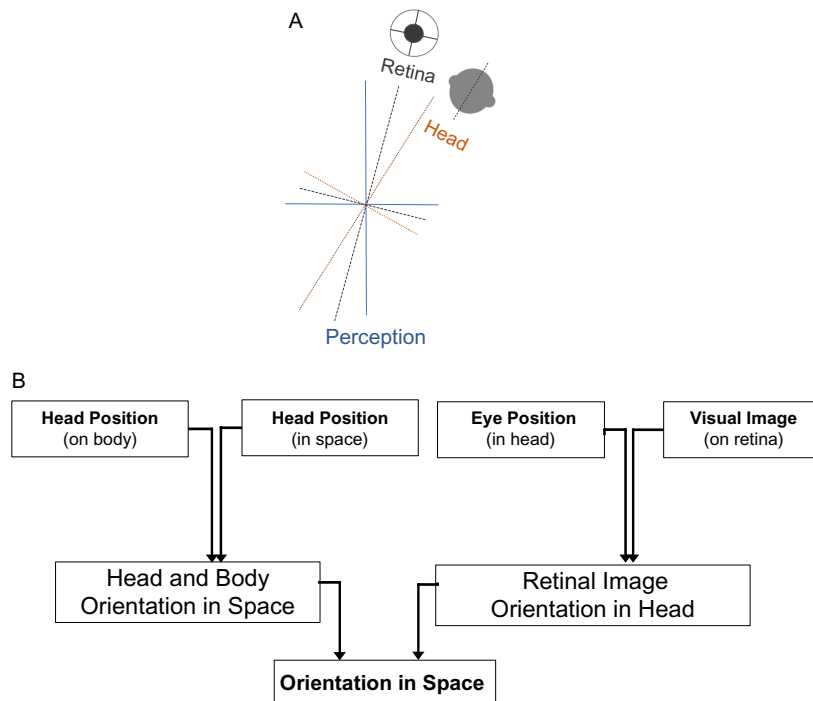


FIG. 1

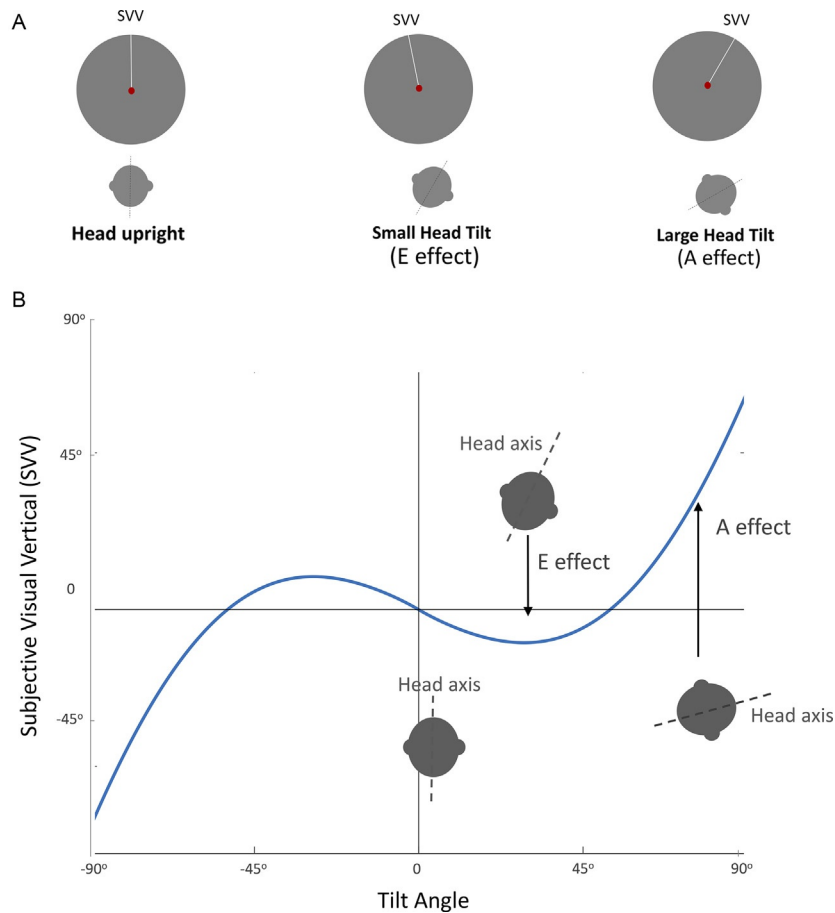
(A) The vestibulo-ocular response during lateral head tilt drives the torsional eye position in the opposite direction of the head tilt. This ocular counter roll (OCR) only partially compensates for the amount of head tilt. As a result, the sensory reference frames for the head, eye and visual world become separated. Despite this separation, the orientation constancy is maintained through multisensory integration for spatial orientation. (B) Perception of spatial orientation involves integration of the visual information with the sensory inputs that encode eye, head, and body positions (e.g., vestibular and proprioceptive inputs).

to study sensory contributions to spatial orientation and probe neural networks involved in this process.

Direction of gravity has been used widely as a reference point to study spatial orientation with respect to the changes in body position (Kheradmand and Winnick, 2017). This is the basis for a psychophysical task known as the subjective visual vertical (SVV) in which a visual line is used to measure perceived earth-vertical orientation in the absence of orientation cues. In the upright position, where the reference frames for the eye, head and visual world are all aligned with the axis of gravity, SVV errors typically remains within two degrees of the earth vertical (Kheradmand and Winnick, 2017; Van Beuzekom and Van Gisbergen, 2000). With a lateral head tilt, however, the partial OCR response (gain ~ 0.10 – 0.25 , driven by the otolith inputs) results in a dissociation of sensory reference frames that encode the head (in space) and eye (in head) positions (Kheradmand and Winnick, 2017; Tarnutzer et al., 2009a,b). Therefore, the brain must integrate inputs represented in disparate sensory reference frames in order to maintain a coherent spatial orientation during lateral head tilts. Such processing demand is reflected by systematic SVV errors, described by studies dating back to 19th century (Fig. 2) (reviewed in Kheradmand and Winnick, 2017). At small head tilt angles ($<60^\circ$), SVV errors are often in the opposite direction of the head tilt, reflecting overestimations of upright orientation relative to the actual head positions (known as the *E*-effect; *E* for “Entgegengesetzt,” German for opposite). At large head tilt angles ($>60^\circ$), SVV errors are usually in the same direction of the head tilt, reflecting underestimations of upright orientation relative to the actual head positions (known as the Aubert or *A*-effect). These systematic errors in spatial orientation do not correspond with the perception of body tilt, which remains fairly accurate at all tilt positions (Bronstein, 1999; Kaptein and Van Gisbergen, 2004; Mast and Jarchow, 1996; Mittelstaedt, 1983; Van Beuzekom et al., 2001; Van Beuzekom and Van Gisbergen, 2000). Such dissociation shows that the spatial orientation is not simply determined by the perception of body position, in line with the multisensory aspect of spatial perception.

2 Bayesian spatial model

A computational model that can effectively address sensory processing and integration for spatial orientation is based on the Bayesian inference principles (Clemens et al., 2011; De Vrijer et al., 2008, 2009; Eggert, 1998; Mac Neilage et al., 2007; Tarnutzer et al., 2009b). The Bayesian spatial model implements an optimal strategy for integration of noisy sensory signals in order to determine upright orientation with respect to changes in the lateral head tilt position. In this framework, sensory signals with less uncertainty (higher signal-to-noise ratio) are given higher weight in the process of multisensory integration (Kheradmand and Winnick, 2017). Thus, the signal reliability (sensory likelihoods) and the chance of encountering a given sensory signal (*prior* probabilities) are taken into account to generate neural estimates that contribute to perception of spatial orientation (Fig. 3). Accordingly,

**FIG. 2**

(A) Subjective visual vertical (SVV) is shown for the head upright, small tilt and large tilt positions. Typically SVV errors are within two degrees of the earth vertical in the upright position. The *E*-effect occurs at small tilt angles with an error deviation in the opposite direction of the tilt. The *A*-effect occurs at large tilt angles with an error deviation in the direction of the tilt.

(B) The SVV errors are shown across tilt positions with the *E*-effect at smaller tilt angles less than 60° and the *A*-effect at tilt angles usually greater than 60°. Negative values indicate leftward or counterclockwise directions and positive values indicate rightward or clockwise directions.

the SVV errors in the model are determined by integration of the sensory signals that encodes head position (i.e., head-in-space position), visual signals from the retina, and sensory signals that encode the torsional position of the eyes (i.e., eye-in-head position). Note that the Bayesian spatial model assumes a vertical orientation of the trunk. Thus, the estimate of head in space is a combination of the otolith sensors and neck proprioception. The *prior* distribution for the head-in-space sensory

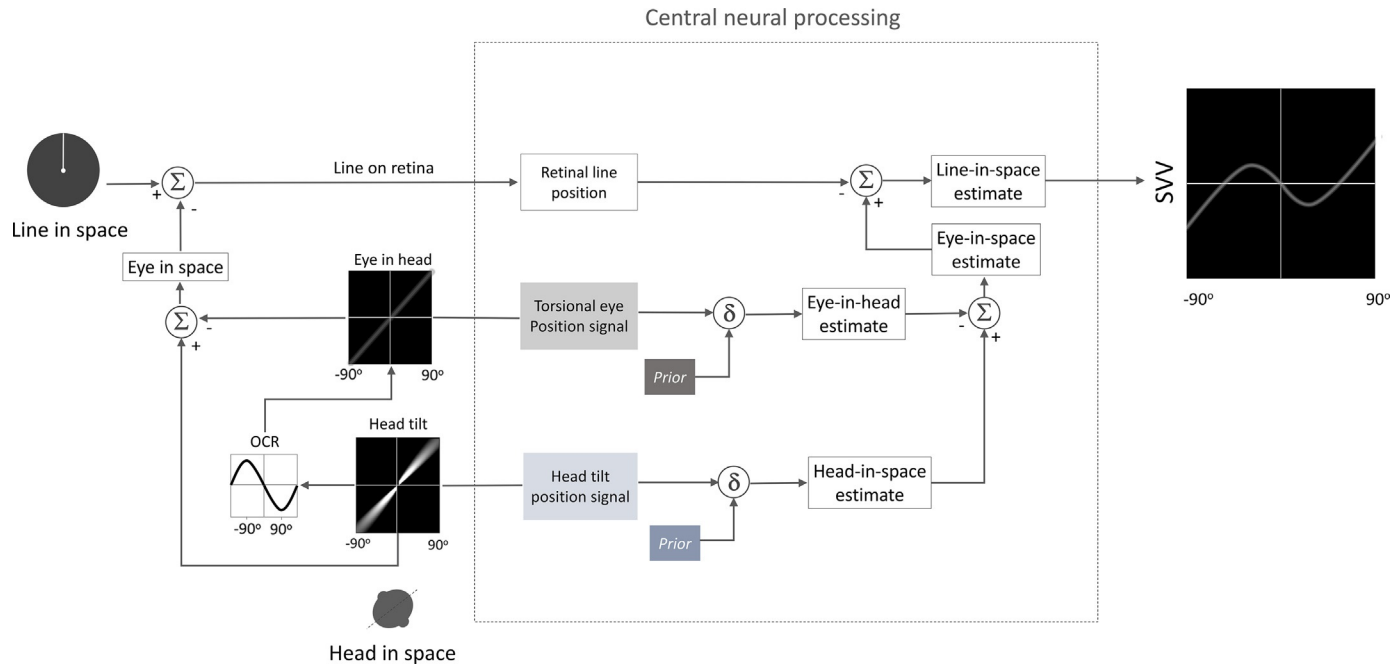


FIG. 3

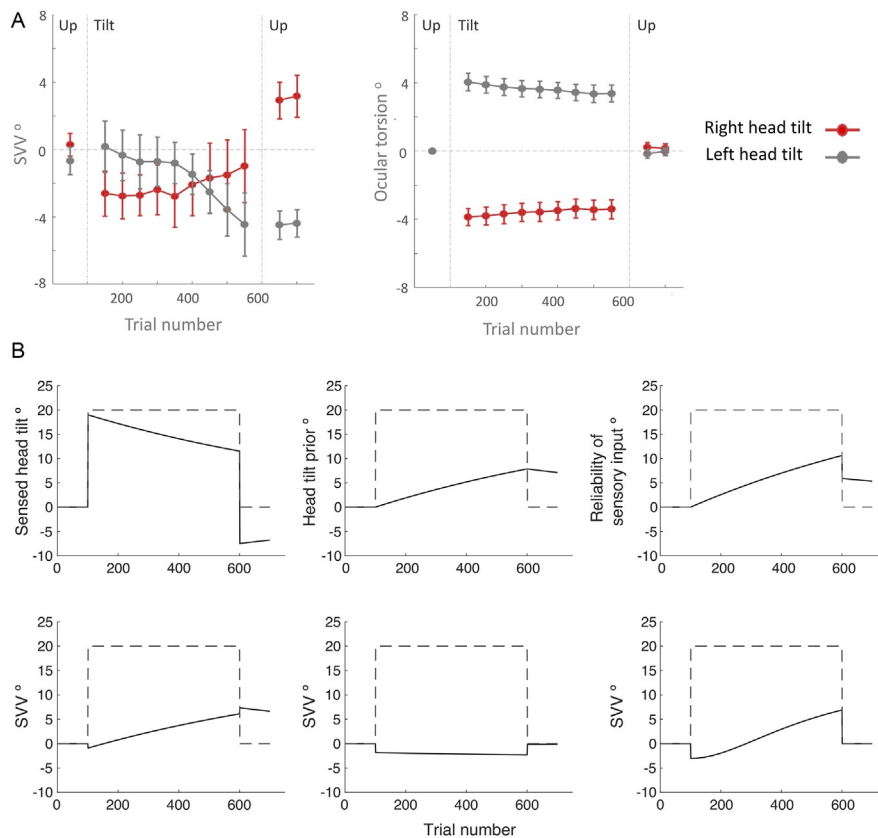
Bayesian model of spatial orientation. A visual stimulus (line in space) is presented in front of a tilted observer ($\pm 90^\circ$ range of tilt position is shown in black graphs). In this model, SW (i.e., the internal estimate of upright orientation) is based on the line-in-space estimate, which is generated by the summation (Σ) of the head-in-space estimate, eye-in-head estimate, and retinal line position. The head tilt position signal is contaminated by Gaussian noise. The eye-in-head position is driven by the ocular counter-roll (OCR), which is contaminated by independent noise. As part of the central neural processing, the head-in-space and eye-in-head estimates are derived by the Bayesian integration (δ) of their sensory signals with their corresponding *priors*.

signals is centered at the earth-vertical orientation as humans spend most of the time in the upright position. This earth-vertical *prior* reduces the effect of noise in the head-in-space sensory signals, but in turn produces a bias in the neural estimate of head tilt position toward the upright position (i.e., the A-effect at large tilt angles). Similarly, the eye-in-head position is integrated with its eye-in-head *prior* distribution to determine the neural estimate of torsional eye position. The torsional eye position signal is a combination of the proprioceptive and efference copy inputs. In this context, a neural underestimation of the eye-in-head position can explain SVV deviation in the opposite direction of the head tilt at small tilt angles (i.e., the E-effect) (De Vrijer et al., 2009).

3 Adaptation to head tilt

Consistent with the multimodal sensory contributions to spatial perception, sensory signals that encode the head, neck and trunk positions have modulating effects on the SVV errors (Alberts et al., 2016; Barbieri et al., 2008; Barra et al., 2010; Day and Wade, 1968, 1969; de Graaf et al., 1992; Guerraz et al., 1998; McKenna et al., 2004; Tarnutzer et al., 2010; Trousselard et al., 2004; Wade, 1968; Young et al., 1975). In this process, there is often a gradual change in perceived spatial orientation during prolonged head tilts, reflected by an SVV drift in the direction of the tilt, and subsequently a post-tilt bias referred to as an SVV aftereffect (Fig. 4A) (Lechner-Steinleitner, 1978; Otero-Millan and Kheradmand, 2016; Tarnutzer et al., 2010, 2013; Wade, 1968, 1970; Wade and Day, 1968a,b). Although, the partial change in ocular torsion during head tilt (i.e., a low OCR gain) is a significant source of SVV errors during lateral head tilts, neither the SVV drift nor the post-tilt aftereffect correlate with the changes in ocular torsion during or after head tilt (Fig. 4A) (Otero-Millan and Kheradmand, 2016). These findings indicate that the otolith signals that drive the torsional eye position during head tilt cannot be the primary source of SVV drift or aftereffect. Similar drifts also occur during haptic measurements of upright perception, which also shows that visual or eye position signals cannot be the source of drift during head tilt (Tarnutzer et al., 2013; Wade and Day, 1968a).

Overall, SVV drifts are often larger and more frequent with the head-on-body tilts compared to the whole-body tilts, which suggest that the drift is primarily driven by the neck proprioception and not the vestibular signals (Day and Wade, 1968; Otero-Millan and Kheradmand, 2016; Tarnutzer et al., 2013; Wade, 1970; Wade and Day, 1968a). In line with these results, the Bayesian spatial model predicts that only a gradual decay in the head-in-space sensory signals can result in the SVV drift and aftereffect in the direction of the head tilt (Fig. 4B) (De Vrijer et al., 2009; Otero-Millan and Kheradmand, 2016). Naturally, during a constant stimulus, sensory inputs reduce over time, and subsequently, an opposite response occurs after the stimulus is removed. According to the Bayesian spatial model, in case of a drift in the head *prior* in the direction of head tilt, the SVV drifts in the opposite, and not in the direction of the head tilt (Fig. 4B). Thus, the SVV drift cannot be explained by the adaptive changes

**FIG. 4**

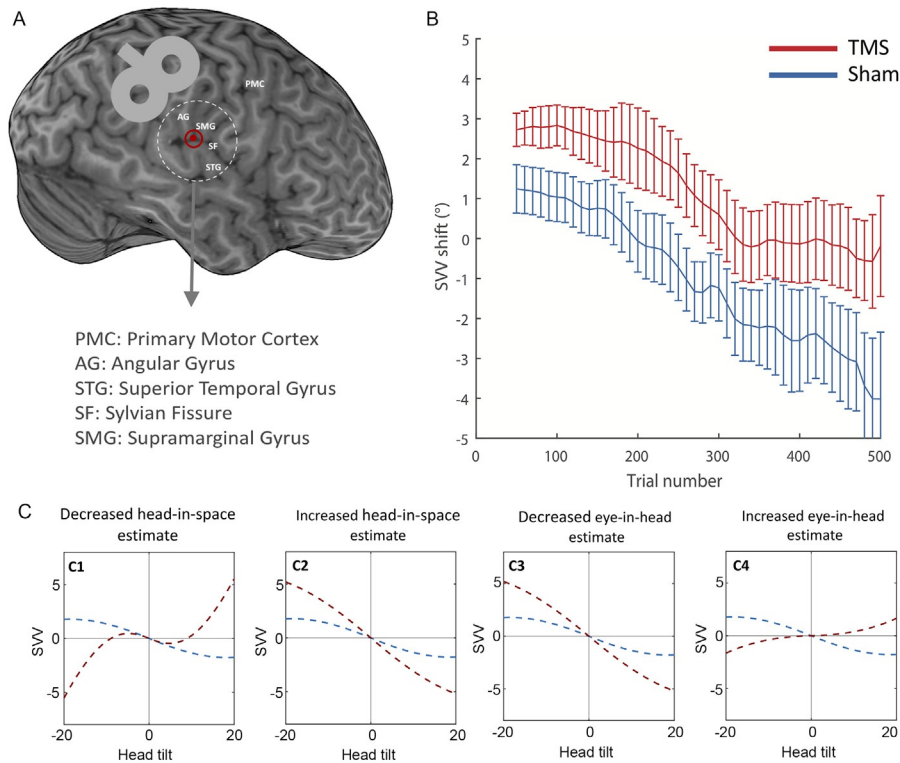
(A) Simultaneous SVV and ocular torsion measurements before, during, and after prolonged head tilts of 20 degrees (Otero-Millan and Kheradmand, 2016). The data points correspond to block of 100 SVV trials and the error bars corresponds to the standard error of mean for 12 subjects. The SVV drift is in the same direction as the head tilt, and when the head returns to upright position there is an aftereffect, also in the same direction as the head tilt. Changes in the ocular torsion do not correspond to the SVV drift or aftereffect. (B) Simulations of the SVV drift and aftereffect by the Bayesian spatial-perception model during and after prolonged 20° head tilt (dashed lines) (Otero-Millan and Kheradmand, 2016). The top panels show changes in the parameters of the model over time and the bottom panels show the SVV drift produced by the model in each scenario (solid lines). In the first column, a decay in the head-in-space sensory inputs during head tilt results in an SVV drift in the direction of the head tilt. After the head is brought back upright, the opposite sensory response can cause an aftereffect. In the second column, a gradual drift in the head *prior*, results in an SVV drift in the opposite direction of the head tilt. In the third column, increasing noise in the head-in-space sensory inputs also results in an SVV drift toward the actual head position. However, when the head is brought back upright, there is no aftereffect as the noise is reduced toward the baseline.

in the head *prior*. Another scenario that can explain SVV drift in the direction of the head tilt is a gradually-increasing noise in the head-in-space sensory signals (i.e., the sensory inputs that encode head tilt position become less reliable over time) (Fig. 4B). In this case, due to a reduced reliability of the head-in-space sensory signals, the *weight* of the head *prior* in the process of head-in-space estimation increases over time, which can result in the SVV drift toward the actual head position. However, when the head is brought back upright, the sensory noise reduces toward the baseline, and it cannot produce the aftereffect. Thus, the noise in the sensory inputs that encode the head position cannot be the source of SVV drift during head tilt or the corresponding post-tilt aftereffect.

4 Neural correlates

Perception of spatial orientation shares the same neural substrate with the perception of body orientation, heading perception, visual gravitational motion perception, sense of embodiment, self-localization and egocentricity (Blanke et al., 2015; Bosco et al., 2008; Cazzato et al., 2015; Donaldson et al., 2015; Hansen et al., 2015; Igelström and Graziano, 2017; Indovina et al., 2005; Jáuregui Renaud, 2015; Kaski et al., 2016; Lacquaniti et al., 2013; Lopez et al., 2008; Roberts et al., 2016; Saj et al., 2014; Silani et al., 2013; Ventre-Dominey, 2014). These “higher order” functions that contribute to sensory processing and integration for spatial perception are localized to the neural networks within the temporo-parietal junction (TPJ). The role of TPJ in spatial orientation is particularly demonstrated in patients with neglect syndrome, with lesions that converge largely at the inferior parietal lobule and posterior aspect of the insular cortex. Patients with TPJ lesions are unable to attend to sensory stimuli in their contralesional hemispace, and they show significant SVV deviations, which are mainly contralesional and biased in the opposite direction of the head tilt (i.e., errors consistent with overestimation of the body tilt position in space) (Baier et al., 2013; Barra et al., 2008, 2010; Bender and Jung, 1948; Bonan et al., 2006; Brandt et al., 1994; Friedmann, 1970; Kerkhoff, 1999; Kerkhoff and Zoelch, 1998; Pérennou et al., 2008; Piscicelli et al., 2015a,b; Saj et al., 2005a; Yelnik et al., 2002). These patients may also have robust contralesional postural deviations that correlate with their errors of spatial orientation (Beevor, 1909; Bjerver and Silfverskiöld, 1968; Bohannon, 1996; Danells et al., 2004; Dieterich and Brandt, 1992; Johannsen et al., 2006a,b; Karnath et al., 2000; Karnath and Broetz, 2003; Paci et al., 2009, 2011; Pérennou et al., 2002, 2008, 2014; Punt and Riddoch, 2002; Saj et al., 2005b; Thömke et al., 2005), which suggests their body is actively aligned with their erroneous spatial orientation (Pérennou et al., 1998, 2002, 2008). These clinical findings highlight the significance of the TPJ network in multisensory integration for spatial orientation (Kheradmand and Winnick, 2017).

The role of TPJ in spatial orientation has also been shown using non-invasive brain stimulation (Kheradmand et al., 2015; Otero-Millan et al., 2018). Accordingly, the disruptive effect of transcranial magnetic stimulation (TMS) at the right TPJ

**FIG. 5**

(A) Transcranial magnetic stimulation (TMS) at the supramarginal gyrus (SMG). (B) SVV results are shown in a groups of nine subjects during 20° left head tilt (Otero-Millan et al., 2018). The average SVV shift from TMS is in the opposite direction of the head tilt, and it is larger than the average SVV shift from a sham stimulation (i.e., a larger *E*-effect with the TMS effect). (C) Simulations by the Bayesian spatial model dissects the effect of TMS (red) versus a normal baseline (blue) on sensory components that contributes to spatial orientation. Since the SVV error is determined by the summation of the head-in-space and eye-in-head neural estimates (Fig. 3), in order to produce SVV errors in the opposite direction of the head tilt, TMS effect should either increase the head-in-space estimate relative to the actual head tilt position (C2) or reduce the eye-in-head estimate relative to the actual torsional eye position (C4).

results in a shift of SVV errors in the opposite direction of the head tilt (Fig. 5A and B). The direction of this error is consistent with the “overestimation” errors found with the TPJ lesions (i.e., increase in *E*-effect). Neither the SVV deviations from TPJ lesions, nor the SVV deviations from the effect of TMS at TPJ are associated with changes in the torsional eye position (Kheradmand et al., 2015; Otero-Millan et al., 2018). These findings suggest that the higher-order neural mechanisms within TPJ

are primarily involved in sensory processing and integration for spatial perception, without having a direct effect on the ocular torsion. Based on these findings, the Bayesian spatial model can be used as a framework to study the computational role of TPJ in sensory integration for spatial orientation (Fig. 5C). According to the model, in order to produce SVV errors in the opposite direction of the head tilt, a neural dysfunction at TPJ should cause an overestimation of the head tilt position or an underestimation of the torsional eye position (Fig. 5C). Since both the head-in-space and eye-in-head neural estimates are derived from their corresponding signals and *prior* distributions, the higher-order mechanisms at TPJ could be involved in processing and integration of these sensory signals into their corresponding neural estimates. Future studies can address these plausible mechanisms by relating the effects of TMS at TPJ to specific processing steps within the Bayesian spatial model.

5 Conclusions

Perception of spatial orientation is generated through integration of sensory information encoding the eye, head and body positions. In this multimodal process, there are systematic errors in spatial orientation with tilting the head or whole body, which reflect challenges for the brain in integrating sensory inputs encoded in disparate sensory reference frames. These errors can be quantified using the SVV task in which perceived earth-vertical orientation is measured in the absence of orientation cues. The computational mechanisms behind these systematic errors are addressed by the Bayesian spatial model that accounts for noisy sensory signals encoding the eye and head positions. In this framework, the reliability of a signal (sensory likelihoods) and the chance of encountering that signal by the brain (*prior* probabilities), together determine a neural estimate of the signal in the process of sensory integration into spatial orientation. Such a Bayesian framework can be applied to examine adaptive changes in spatial orientation during head tilt and study neural substrate that contribute to perception of spatial orientation.

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